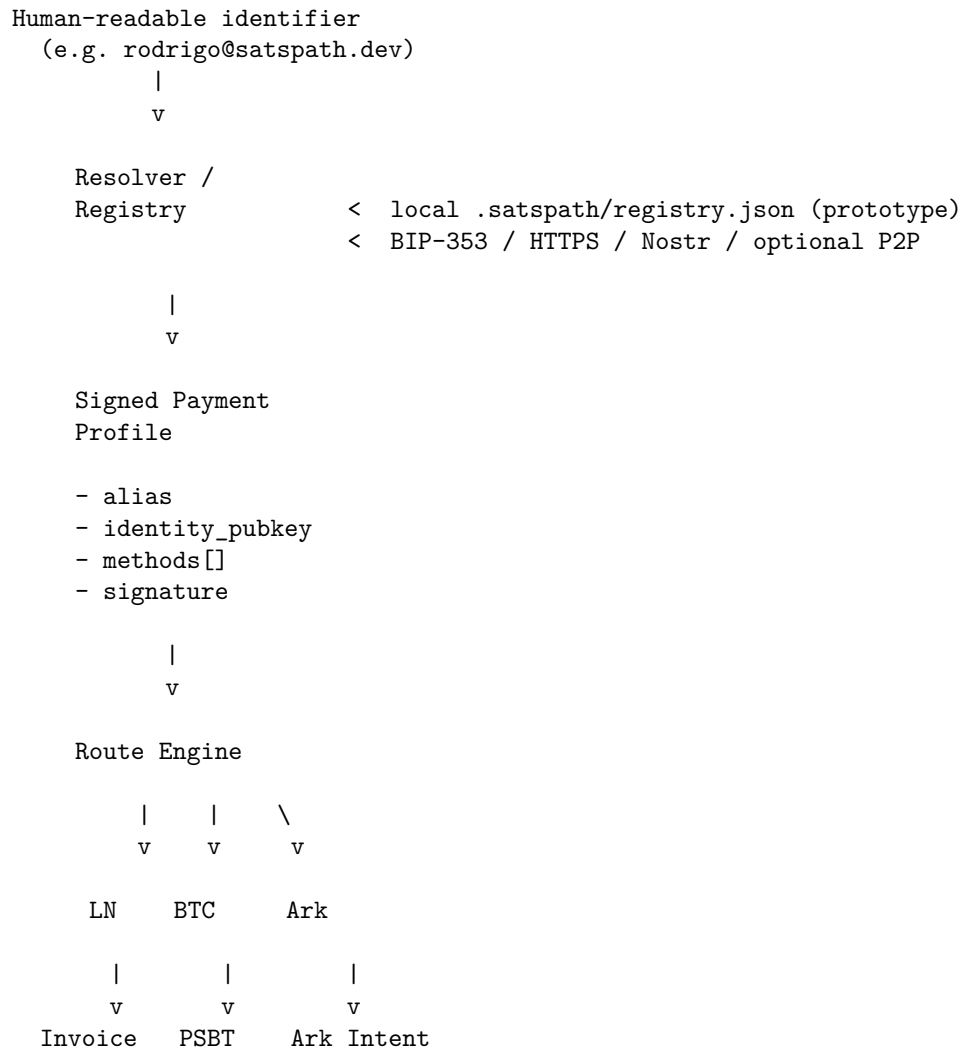


SatsPath Architecture

Overview

SatsPath is a routing and resolution layer that sits above existing Bitcoin payment protocols. It does not replace Lightning, Ark, or on-chain payments — it resolves human-readable identifiers to signed payment profiles and selects the best available rail.

System Diagram



Crate Structure

```
satspath/
  crates/
    satspath-core/
      profile.rs    # Types, crypto, codec, registry
      crypto.rs     # PaymentProfile, PaymentMethod, SignedPaymentProfile
      codec.rs      # secp256k1 keypair, sign, verify, fingerprint
      registry.rs   # encode/decode satspath: URIs
      errors.rs     # local file registry
                   # SatsPathError enum
```

```

satspath-router/      # Route selection engine
  router.rs           # select_route() - priority logic
  fees.rs             # mempool.space fee API client
  lightning.rs        # Lightning availability + fee helpers
  onchain.rs          # On-chain fee math + availability
  ark.rs              # ArkClient trait + MockArkClient

satspath-cli/         # User-facing binary
  commands/           # One module per CLI subcommand

```

Data Flow

Payment request lifecycle

1. Sender knows: rodrigo@satspath.dev + 21000 sats
2. Encode (optional):
`satspath:v1:<base64url({"version":1,"alias":"rodrigo@satspath.dev","amount_sats":21000,...})>`
3. Decode:
`PaymentRequest { alias, amount_sats, memo, ... }`
4. Resolve:
`Registry::resolve_alias("rodrigo@satspath.dev") -> SignedPaymentProfile`
5. Verify:
`verify_signed_profile(signed) -> bool`
6. Route:
`select_route(RouteRequest { alias, amount_sats, signed_profile })`
`-> RouteQuote { selected_method, reason, fee, confirmation }`
7. Pay (simulated):
Execute against selected rail (Lightning invoice, on-chain tx, Ark intent)

Invite lifecycle (unregistered receiver)

1. Sender tries to pay julian@example.com
2. Registry returns AliasNotFound
3. `create_invite("julian@example.com", amount_sats) -> Invite`
4. Invite contains alias_hash, claim_url, warning
5. Sender shares claim_url with receiver out-of-band
6. Receiver generates their own keys locally and registers
7. Sender retries payment

Security Boundaries

- Private keys never leave the user's machine.
- The registry stores only public data (pubkeys, addresses, signatures).
- Signature verification happens on the receiver side before any payment.
- The `.satspath/` directory is git-ignored.

Pluggability

The registry is an abstraction — in production, `Registry::open()` can be swapped for:

- BIP-353 DNS TXT record lookup
- Nostr NIP-05 / NIP-57 resolution
- Lightning Address `.well-known/lnurlp` discovery
- A decentralized DHT

The router is similarly pluggable: new payment rails are added by implementing the routing priority logic and returning a `RouteQuote`.